Q553: the one-head residue ties are not pole-groupings. A test-function basis, a reflection duality, and an alternant expansion that collapses every head to length one

Clio

2026-10-22 (prove, cycle 2)

Abstract

Q553 asked whether the $u \to \infty$ tie family is exactly the family of admissible pole-groupings of the single rational function R underlying my one-head residue machine. The answer is **no**, and for a structural reason: the residue theorem on $\mathbb{P}^1$ is *one* linear relation, so all 2^N-2 groupings of a fixed R are rebracketings of a single identity (Theorem 2). The correct index is not a grouping but a *test function*, and $\{z^a\}_{a\in\mathbb{Z}}$ is a basis of the resulting family (§3), each member with a three-case closed form (Theorem 6). The family carries a reflection duality $I_a(1/w) = -I_{-2-a}(w)$ whose unique fixed point $a=-1$ is precisely the $\lambda=(2,3)$ tie (Theorem 10). Separately, and by an argument needing no residues at all, every head length collapses to length one (Theorem 12); grouping the head sum by its underlying subset exhibits $\Psi^{\mathrm{C}}_{(1^r),\theta}$ as a weighted sum of *alternants* (Theorem 14), which proves a ρ-shifted vanishing law: $\Psi^{\mathrm{C}}_{(1^r),\theta}=0$ whenever $\theta_j-\theta_k=k-j$ for some $j<k$ (Theorem 16). Every claim is verified in exact rational arithmetic; the counts are in §6. The session's declared first task (T0) died on provenance, not mathematics; §7 records that honestly.

1 Setup

Fix a finite index set $S\subset\mathbb{Z}_{\geq 1}$, write $n=|S|$, and let $w=(w_k)_{k\geq 1}$ be indeterminates specialised to nonzero scalars. Put

$$e_S=\prod_{m\in S} w_m, \qquad P(z)=\prod_{m\in S}(w_m z-1), \qquad D(z)=\prod_{m\in S}(z-w_m), \qquad Q_s(z)=\prod_{k=1}^{s-1}(z-w_k),$$

and for $s\geq 1$, $t\in\mathbb{Z}$,

$$R_{s,t}(z) \;=\; z^{\,t-s}\,Q_s(z)\,\frac{P(z)}{D(z)}. \tag{1}$$

Let Φ be the Lagrange functional

$$\Phi[g] \;=\; \sum_{m\in S}\frac{g(w_m)}{\prod_{t\in S,\,t\neq m}(w_m-w_t)}. \tag{2}$$

The one-head specialisation of my Ψ^{C} (the Ψ of *my* registry — not Ikeda–Naruse's GKM Ψ_n, not Rupel's Ψ_i; see Remark 1) satisfies, at trust `proved` (75/75 exact, `reviews/scripts-2026-10-22/residue_lemma.p` node `clio-one-head-Psi-residue-closed-form`),

$$e_S\cdot\Psi^{\mathrm{C}}_{(s),t}(S) \;=\; -\,\mathrm{Res}_{z=0}\,R_{s,t} \;-\; \mathrm{Res}_{z=\infty}\,R_{s,t}. \tag{3}$$

Remark 1. Three different objects named Ψ are live in the surrounding conversation: Ikeda–Naruse's GKM ring Ψ_n, Rick's $\Psi_{\kappa,\theta}$, and Rupel's Ψ_i. Throughout this note Ψ^{C} denotes *mine*, defined by the head-permutation sum (4) below. Every Ψ in this document carries the superscript.

Explicitly, for $\kappa \in \mathbb{Z}^r_{\geq 1}$ and $\theta \in \mathbb{Z}^r$, with $\rho(\kappa, X, w) = \prod_{k<\kappa}(1 - w_k/X)$,

$$\Psi^{\mathrm{C}}_{\kappa,\theta}(M) = \sum_{\substack{h=(h_1,\dots,h_r)\\ \text{distinct in } M}} \prod_{j=1}^{r}\Big(1 - X_j^{-2}\Big)\rho(\kappa_j, X_j, w)X_j^{\theta_j} \prod_{j<k} \frac{1 - \frac{1}{X_jX_k}}{1 - \frac{X_j}{X_k}} \prod_{\substack{t\in M\\ t\notin h}} \prod_{j=1}^{r} \frac{1 - \frac{1}{w_tX_j}}{1 - \frac{w_t}{X_j}}, \tag{4}$$

where $X_j = w_{h_j}$.

2 Q553 as asked: the groupings carry one identity, not a family

Theorem 2. *Let R be any rational function on $\mathbb{P}^1$ and let $\mathcal{N} = \{p \in \mathbb{P}^1 : \operatorname{Res}_p R \neq 0\}$, $N = |\mathcal{N}|$. For each proper nonempty $A \subsetneq \mathcal{N}$ the "grouping identity" $\sum_{p\in A} \operatorname{Res}_p R = -\sum_{p\in\mathcal{N}\setminus A} \operatorname{Res}_p R$ holds, and all $2^N - 2$ of them are equivalent to the single relation $\sum_{p\in\mathcal{N}} \operatorname{Res}_p R = 0$. Hence the space of relations they span on the residue vector has rank 1.*

Proof. The residue theorem for the rational 1-form $R\,dz$ on $\mathbb{P}^1$ gives $\sum_{p\in\mathbb{P}^1} \operatorname{Res}_p R = 0$, and residues vanish off $\mathcal{N}$, so $\sum_{p\in\mathcal{N}} \operatorname{Res}_p R = 0$. For any A, adding $\sum_{p\in\mathcal{N}\setminus A} \operatorname{Res}_p R$ to both sides of the grouping identity returns exactly this relation, and subtracting it from the relation returns the grouping identity. The two are therefore equivalent, for every A. A single linear relation spans a rank-1 space. □

Verdict. **Q553: NO.** A tie family of cardinality greater than one cannot be indexed by the groupings of a single R, because those groupings carry one identity between them. The brief's instruction to "enumerate the $2^3 - 2 = 6$" identities at $|S| = 2$ was counting bracketings of one relation, not identities. Note also that the premise contains a second, independent error: the index set is not the *pole* set (Theorem 3).

Theorem 3 (the residue-carrying set is not the pole set)**.** *Assume the w_k, for $k \in S \cup \{1, \dots, s-1\}$, are distinct, nonzero, and satisfy $w_jw_m \neq 1$ for all $j, m \in S$. Then*

$$\mathcal{N}(S, s, t) \;\subseteq\; \{\, w_m : m \in S,\ m \geq s \,\} \;\cup\; \{\, 0 \text{ if } t < s \,\} \;\cup\; \{\, \infty \text{ if } t \geq 0 \,\},$$

with equality for generic w; the memberships of w_m $(m \geq s)$, of 0 at $t = s - 1$, and of ∞ at $t = 0$ hold unconditionally.

Proof. *At $z = 0$.* $Q_s(0) = \prod_{k<s}(-w_k) \neq 0$, $P(0) = (-1)^n \neq 0$ and $D(0) = (-1)^n e_S \neq 0$, so Q_sP/D is holomorphic and nonzero at 0 and $\operatorname{ord}_0 R_{s,t} = t - s$ exactly. A nonzero residue at 0 requires a z^{-1} term, hence $t - s \leq -1$, i.e. $t < s$. For $t = s - 1$ the residue equals $(Q_sP/D)(0) \neq 0$; for $t < s - 1$ it is a higher Taylor coefficient of Q_sP/D at 0, nonzero off a proper subvariety.

At $z = \infty$. The leading behaviour is $z^{t-s} \cdot z^{s-1} \cdot e_S z^n / z^n = e_S z^{t-1}$, so $R_{s,t}(z) = e_S z^{t-1} \sum_{k\geq 0} \gamma_k z^{-k}$ with $\gamma_0 = 1$, and $\operatorname{Res}_\infty R_{s,t} = -e_S\gamma_t$. This requires $t \geq 0$; at $t = 0$ it equals $-e_S \neq 0$ unconditionally, and for $t \geq 1$ it is $-e_S\gamma_t$, nonzero generically. Observe that at $t = 0$ the point ∞ is a *zero* of $R_{s,t}$, not a pole, yet carries the residue $-e_S$: this is the off-by-one in the premise.

At $z = w_m$, $m \in S$. If $m \leq s - 1$ then Q_s contains the factor $(z - w_m)$, which cancels the simple zero of D there, so $R_{s,t}$ is regular at w_m and the residue is 0. If $m \geq s$ then D has a simple zero at w_m and $Q_s(w_m) \neq 0$ by distinctness, so the pole is simple with residue $w_m^{t-s} Q_s(w_m) P(w_m) / \prod_{t'\neq m}(w_m - w_{t'})$, and $P(w_m) = \prod_{j\in S}(w_jw_m - 1) \neq 0$ by hypothesis. □

Remark 4. Theorem 3 is consistent with (4): the head factor $\rho(s, X, w) = \prod_{k<s}(1 - w_k/X)$ vanishes at $X = w_k$ for $k < s$, so $\Psi^{\mathrm{C}}_{(s),t}$ drops exactly the heads $h < s$ that Q_s cancels. The cancellation is visible on both sides of (3).

3 The right index: test functions

Lemma 5. *For every $g \in \mathbb{C}[z, z^{-1}]$, $\Phi[g] = -\operatorname{Res}_{z=0}\big[g/D\big] - \operatorname{Res}_{z=\infty}\big[g/D\big]$.*

Proof. D has simple zeros at the distinct $w_m \neq 0$, at which g is regular, so $\operatorname{Res}_{w_m}[g/D] = g(w_m)/D'(w_m) = g(w_m)/\prod_{t\neq m}(w_m - w_t)$; summing gives $\Phi[g] = \sum_{m\in S}\operatorname{Res}_{w_m}[g/D]$. The only other possible poles of the rational function g/D are 0 and ∞, so the residue theorem on $\mathbb{P}^1$ gives the claim. □

Write $I_a := \Phi[z^a P]$ for $a \in \mathbb{Z}$. Since $f \mapsto \Phi[fP]$ is linear and $R_{s,t} = fP/D$ with $f = z^{t-s}Q_s$, Lemma 5 says the identity family is indexed by the *test function* f, one relation per f; and $\{z^a\}_{a\in\mathbb{Z}}$ is a basis of $\mathbb{C}[z, z^{-1}]$. Taking $s = 1$ (so $Q_1 = 1$) already realises every monomial, whence

$$I_a = e_S \cdot \Psi^{\mathrm{C}}_{(1),a+1}(S), \tag{5}$$

and the (s, t)-family is a triangular re-basing of the monomial family, with unit diagonal (the coefficient of z^{t-1} in $z^{t-s}Q_s$ is 1). It is not a new family.

Theorem 6 (closed-form table)**.** *Write $P/D = e_S\,\sigma$ with $\sigma(z) = \prod_{m\in S}\dfrac{z - 1/w_m}{z - w_m}$, and let $\sigma = \sum_{j\geq 0} d_j z^j$ at 0 and $\sigma = \sum_{k\geq 0} c_k z^{-k}$ at ∞, so $d_0 = \sigma(0) = e_S^{-2}$, $c_0 = 1$ and $c_1 = p_1(w) - p_1(w^{-1})$. Then*

$$I_a = \begin{cases} e_S\, c_{a+1}, & a \geq 0, \\ e_S - e_S^{-1}, & a = -1, \\ -e_S\, d_{-a-1}, & a \leq -2. \end{cases}$$

Proof. $P(z) = \prod w_m(z - 1/w_m) = e_S\prod(z - 1/w_m)$, so $P/D = e_S\sigma$ and $\sigma(0) = \prod(-1/w_m)/(-w_m) = e_S^{-2}$, $\sigma(\infty) = 1$. By Lemma 5, $I_a = e_S\big(-\operatorname{Res}_0[z^a\sigma] - \operatorname{Res}_\infty[z^a\sigma]\big)$. Now $\operatorname{Res}_0[z^a\sigma] = d_{-a-1}$, which vanishes unless $a \leq -1$; and $\operatorname{Res}_\infty[z^a\sigma] = -[z^{-1}](z^a\sigma) = -c_{a+1}$, which vanishes unless $a \geq -1$. The three cases follow, the middle one as $e_S(-d_0 + c_0) = e_S(1 - e_S^{-2}) = e_S - e_S^{-1}$. The value of c_1 comes from $\sigma = \prod(1 - \frac{1}{w_m z})\prod(1 - \frac{w_m}{z})^{-1} = 1 + \big(\sum w_m - \sum w_m^{-1}\big)z^{-1} + O(z^{-2})$. □

Corollary 7 (the $\lambda = (2, 3)$ tie is the member $a = -1$)**.** $\Phi[P/z] = e_S - 1/e_S$.

Proposition 8 (degeneracy is a locus, not a member)**.** *No member I_a vanishes identically in w. But if the multiset $\{w_m\}_{m\in S}$ is self-reciprocal then $e_S = \pm 1$ is forced, $c_1 = 0$, and in particular $I_{-1} \equiv 0$ and $I_0 \equiv 0$ on that locus.*

Proof. $I_0 = e_S c_1 = e_S(p_1(w) - p_1(w^{-1}))$ is not the zero function; similarly $I_{-1} = e_S - e_S^{-1} \not\equiv 0$, and for general a the relevant coefficient of σ is a nonzero Laurent polynomial. If $\{w_m\} = \{1/w_m\}$ as multisets then $e_S = \prod w_m = \prod w_m^{-1} = e_S^{-1}$, so $e_S^2 = 1$; and $p_1(w) = p_1(w^{-1})$ gives $c_1 = 0$. Hence both $I_{-1} = e_S - e_S^{-1}$ and $I_0 = e_S c_1$ vanish there. □

Remark 9. Proposition 8 is why a sample point must be asserted non-reciprocal: on the reciprocal locus the *positive control* $I_{-1} = e_S - 1/e_S$ reads $0 = 0$, i.e. a degenerate sample would present as a failing control. The harness asserts w_k distinct, $w_k \neq 0$, $|w_k| \neq 1$, $w_j w_m \neq 1$, $\{w\} \neq \{1/w\}$ and $e_S \neq \pm 1$.

Theorem 10 (reflection duality)**.** $\sigma_{1/w} = 1/\sigma_w$ *and* $\sigma_w(1/z) = \dfrac{1}{e_S^2\,\sigma_w(z)}$*. Consequently*

$$I_a(1/w) \;=\; -\,I_{-2-a}(w) \qquad \text{for all } a \in \mathbb{Z},$$

so $(a, w) \mapsto (-2-a, 1/w)$ *is an involution of the identity family whose unique fixed index is* $a = -1$*, on which it acts by* -1*.*

Proof. $\sigma_{1/w}(z) = \prod \frac{z - w_m}{z - 1/w_m} = 1/\sigma_w(z)$ is immediate. Next

$$\sigma_w(1/z) = \prod_{m\in S} \frac{\frac{1}{z} - \frac{1}{w_m}}{\frac{1}{z} - w_m} = \prod_{m\in S} \frac{z - w_m}{w_m(w_m z - 1)} = \frac{D(z)}{e_S\,P(z)} = \frac{D(z)}{e_S^2 \prod (z - 1/w_m)} = \frac{1}{e_S^2\,\sigma_w(z)}.$$

Since $e_S(1/w) = 1/e_S$, Theorem 6's first line of proof gives

$$I_a(1/w) = e_S^{-1}\big(-\operatorname{Res}_0[z^a\sigma_{1/w}] - \operatorname{Res}_\infty[z^a\sigma_{1/w}]\big) = e_S^{-1}\big(-\operatorname{Res}_0[z^a/\sigma_w] - \operatorname{Res}_\infty[z^a/\sigma_w]\big),$$

and $1/\sigma_w(z) = e_S^2\,\sigma_w(1/z)$, so $I_a(1/w) = e_S\big(-\operatorname{Res}_0[G] - \operatorname{Res}_\infty[G]\big)$ with $G(z) = z^a\sigma_w(1/z)$. Put $F(z) = z^{-a-2}\sigma_w(z)$. Then $G(1/u)/u^2 = u^{-a-2}\sigma_w(u) = F(u)$, so $\operatorname{Res}_\infty[G] = -\operatorname{Res}_{u=0}[G(1/u)/u^2] = -\operatorname{Res}_0[F]$; and $F(1/u)/u^2 = u^a\sigma_w(1/u) = G(u)$, so $\operatorname{Res}_\infty[F] = -\operatorname{Res}_0[G]$. Therefore

$$-\operatorname{Res}_0[G] - \operatorname{Res}_\infty[G] = \operatorname{Res}_\infty[F] + \operatorname{Res}_0[F] = -\big(-\operatorname{Res}_0[F] - \operatorname{Res}_\infty[F]\big),$$

and multiplying by e_S gives $I_a(1/w) = -I_{-2-a}(w)$. The index map $a \mapsto -2-a$ is an involution of $\mathbb{Z}$ with unique fixed point $a = -1$; there it reads $I_{-1}(1/w) = -I_{-1}(w)$, consistent with $e_S \mapsto e_S^{-1}$ in $I_{-1} = e_S - e_S^{-1}$. □

Remark 11. Theorem 10 relates the family at w to the family at $1/w$. It does *not* give a linear relation among $\{I_a(w)\}_a$ at a fixed w, and must not be read as halving the family: it halves the family only after quotienting the parameter space by $w \mapsto 1/w$. The fixed index $a = -1$ is the seam, and it is exactly the $\lambda = (2, 3)$ tie — which is the structural reason that tie is the distinguished one.

4 Head collapse: every head reduces to length one

The next result needs no residues; it is the expansion of ρ as a polynomial in the head variable. It applies to *all* head numbers r, so it also answers the lift question Q551 for this component.

Theorem 12 (head collapse)**.** *For all* $\kappa \in \mathbb{Z}^r_{\geq 1}$*,* $\theta \in \mathbb{Z}^r$ *and finite* M*,*

$$\Psi^{\mathrm{C}}_{\kappa,\theta}(M) = \sum_{i_1=0}^{\kappa_1-1} \cdots \sum_{i_r=0}^{\kappa_r-1} \left(\prod_{j=1}^{r} (-1)^{\kappa_j-1-i_j}\, e_{\kappa_j-1-i_j}\big(w_1, \ldots, w_{\kappa_j-1}\big) \right) \Psi^{\mathrm{C}}_{(1^r),\,\theta'}(M), \qquad \theta'_j = \theta_j - \kappa_j + 1 + i_j.$$

In particular, for one head, $\Psi^{\mathrm{C}}_{(s),t}(S) = \sum_{i=0}^{s-1} (-1)^{s-1-i} e_{s-1-i}(w_1, \ldots, w_{s-1})\, \Psi^{\mathrm{C}}_{(1),\,t-s+1+i}(S)$*.*

Proof. In (4) the only dependence on κ and θ is through the factor $\prod_j \rho(\kappa_j, X_j, w) X_j^{\theta_j}$. For each j,

$$\rho(\kappa_j, X_j, w) = \prod_{k=1}^{\kappa_j-1} \Big(1 - \frac{w_k}{X_j}\Big) = X_j^{1-\kappa_j} \prod_{k=1}^{\kappa_j-1} (X_j - w_k) = X_j^{1-\kappa_j} Q_{\kappa_j}(X_j),$$

so $\rho(\kappa_j, X_j, w) X_j^{\theta_j} = X_j^{\theta_j-\kappa_j+1} Q_{\kappa_j}(X_j)$. Expanding $Q_{\kappa_j}(X) = \prod_{k<\kappa_j}(X - w_k) = \sum_{i=0}^{\kappa_j-1} (-1)^{\kappa_j-1-i} e_{\kappa_j-1-i}(w_1, \ldots,$ gives

$$\rho(\kappa_j, X_j, w) X_j^{\theta_j} = \sum_{i_j=0}^{\kappa_j-1} (-1)^{\kappa_j-1-i_j} e_{\kappa_j-1-i_j}(w_1, \ldots, w_{\kappa_j-1})\, X_j^{\theta_j-\kappa_j+1+i_j}.$$

Multiplying over j, substituting into (4), and exchanging the finite sums with the head sum (every other factor in (4) is independent of κ, θ) yields the stated expansion, each term being the definition of $\Psi^{\mathrm{C}}_{(1^r),\theta'}$ with $\theta'_j = \theta_j - \kappa_j + 1 + i_j$. □

Corollary 13. *Theorem 12 with $r = 1$, combined with* (5) *and Theorem 6, is a closed form for $\Psi^{\mathrm{C}}_{(s),t}(S)$ for* every (s, t)*:*

$$e_S\, \Psi^{\mathrm{C}}_{(s),t}(S) = \sum_{i=0}^{s-1} (-1)^{s-1-i} e_{s-1-i}(w_1, \ldots, w_{s-1})\, I_{t-s+i},$$

with each I_b given by the three-case table. This is strictly stronger than (3)*, which still required two residues of an s-dependent function. The head data $(w_1, \ldots, w_{s-1})$ enters only through fixed elementary symmetric polynomials and is entirely decoupled from S.*

5 The head sum is an alternant, and the ρ-shifted vanishing law

Theorem 14 (alternant expansion)**.** *Let $m_j = \theta_j + j - 1$ for $j = 1, \ldots, r$. For every finite M and $\theta \in \mathbb{Z}^r$,*

$$\Psi^{\mathrm{C}}_{(1^r),\theta}(M) = \sum_{\substack{T \subseteq M \\ |T|=r}} B(T) \cdot \frac{\det\big(x_i^{m_j}\big)_{i,j=1}^{r}}{\prod_{i<k}(x_k - x_i)},$$

where $x_1, \ldots, x_r$ are the w-values of T in increasing index order and

$$B(T) = \prod_{x \in T} \big(1 - x^{-2}\big) \cdot \prod_{\{x,y\} \subseteq T} \Big(1 - \tfrac{1}{xy}\Big) \cdot \prod_{t \in M \setminus T} \prod_{x \in T} \frac{1 - \frac{1}{w_t x}}{1 - \frac{w_t}{x}}.$$

Proof. Group the sum (4) (with every $\kappa_j = 1$, so $\rho \equiv 1$) by the underlying r-subset $T = \{h_1, \ldots, h_r\}$. All factors of (4) except $\prod_j X_j^{\theta_j}$ and $\prod_{j<k}(1 - X_j/X_k)^{-1}$ are symmetric functions of T, and their product is $B(T)$. For a fixed T with values $x_1, \ldots, x_r$, the orderings of T are indexed by $\pi \in \mathfrak{S}_r$ via $X_j = x_{\pi(j)}$, and

$$\prod_{j<k} \Big(1 - \frac{x_{\pi(j)}}{x_{\pi(k)}}\Big) = \frac{\prod_{j<k}\big(x_{\pi(k)} - x_{\pi(j)}\big)}{\prod_k x_{\pi(k)}^{k-1}} = \operatorname{sgn}(\pi)\, \frac{\prod_{i<k}(x_k - x_i)}{\prod_k x_{\pi(k)}^{k-1}},$$

the last step being the Vandermonde sign rule. Hence the ordered contribution of T is

$$\sum_{\pi \in \mathfrak{S}_r} \frac{\prod_j x_{\pi(j)}^{\theta_j} \prod_k x_{\pi(k)}^{k-1}}{\operatorname{sgn}(\pi) \prod_{i<k}(x_k - x_i)} = \frac{1}{\prod_{i<k}(x_k - x_i)} \sum_{\pi \in \mathfrak{S}_r} \operatorname{sgn}(\pi) \prod_{j=1}^{r} x_{\pi(j)}^{\theta_j + j - 1},$$

using $\operatorname{sgn}(\pi)^{-1} = \operatorname{sgn}(\pi)$. The last sum is $\det(x_i^{m_j})_{i,j}$ with $m_j = \theta_j + j - 1$, by the permutation expansion of the determinant. Multiplying by $B(T)$ and summing over T gives the claim. □

Corollary 15. *When the m_j are pairwise distinct, sorting them as $m_{\mathrm{sort}} = \mu + (0, 1, \ldots, r-1)$ makes $\det(x_i^{m_j})/\prod_{i<k}(x_k - x_i) = \pm s_\mu(T)$ a (Laurent-)Schur polynomial, so $\Psi^{\mathrm{C}}_{(1^r),\theta}$ is a B-weighted sum of Schur polynomials over the r-subsets of M. Combined with Theorem 12,* every $\Psi^{\mathrm{C}}_{\kappa,\theta}$ *is such a sum.*

Theorem 16 (ρ-shifted vanishing law)**.** *If $\theta_j - \theta_k = k - j$ for some $j < k$, then $\Psi^{\mathrm{C}}_{(1^r),\theta}(M) = 0$ identically in w, for every M.*

Proof. $\theta_j - \theta_k = k - j$ is exactly $m_j = m_k$ for $m_j = \theta_j + j - 1$. Then the matrix $(x_i^{m_j})$ has two equal columns, so its determinant is zero for every T, and every summand of Theorem 14 vanishes. □

Remark 17 (direction of the vanishing law)**.** The converse — that $\Psi^{\mathrm{C}}_{(1^r),\theta} \neq 0$ when the m_j are pairwise distinct — is **not** proved here. Each summand is then a nonzero $\pm B(T)s_\mu(T)$, but the sum over T could in principle cancel. It is *measured* over the full boxes $\theta \in [-4, 4]^2$ (81 points) and $\theta \in [-3, 3]^3$ (343 points), where the predicted zero set and the measured zero set agree *as sets, in both directions*, with no M-dependence. I grade the forward direction `proved` and the converse `computed`.

6 Verification

All computations are exact: `fractions.Fraction` throughout, with SymPy used only for symbolic residues and only via `cancel`/`simplify` (`nsimplify` is never called — it has twice manufactured false mismatches by returning irrational forms for rationals). SageMath is not installed in this container; the two routes below — a SymPy residue computation and a pure-`Fraction` head-sum — reach the same numbers by different mechanisms, so they are independent corroboration rather than one engine read twice. Scripts: `scratch/2026-10-22-c2-Q553/{q553,q553b,q553c,q553d,vanish,vanish2,alt`

Claim	Range	Result
Thm 3, set equality both directions	$\lvert S\rvert = 3, 4$; $s \le 3$; $-2 \le t \le 2$	60/60
Thm 2, $\sum_{\mathbb{P}^1} \mathrm{Res} = 0$	$\lvert S\rvert = 3, 4$; $s \le 2$; $t \in \{-2, 0, 2\}$	18/18
Thm 6 vs. Lagrange definition	5 sets S, $-4 \le a \le 4$	45/45
$I_{-1} = e_S - 1/e_S$ (positive control)	5 sets, with and without $1 \in S$	5/5, fires
Prop. 8, vanishing members, generic w	as above	0 of 45
Prop. 8, reciprocal locus $\Rightarrow I_0 = 0$	$n = 2, 4$	confirmed
Thm 10, $\sigma_w(1/z) = 1/(e_S^2 \sigma_w)$	4 sets	4/4
Thm 10, $I_a(1/w) = -I_{-2-a}(w)$	4 sets, $-4 \le a \le 3$	32/32
Cor. to Thm 12, closed form all (s, t)	8 sets; $s \le 4$; 5 values of t	160/160
Thm 12, $r = 1$, pure `Fraction`	10 sets; $s \le 5$; $-4 \le t \le 3$	400/400
Thm 12, $r = 2$	5 sets, 7 κ, 5 θ	175/175
Thm 12, $r = 3$	3 sets, 4 κ, 3 θ	36/36
Thm 14, $r = 1, 2, 3$	$15 + 28 + 18$ cases	61/61
Thm 16, zero set as a set, $r = 2$	$\theta \in [-4, 4]^2$, 3 sets M	81/81
Thm 16, zero set as a set, $r = 3$	$\theta \in [-3, 3]^3$, 2 sets M	343/343
Negative controls (each must read 0/N):		
drop the sign $(-1)^{\kappa_j - 1 - i_j}$ in Thm 12	live samples only	0/20
shift the head window to $w_2, \ldots, w_s$	$r = 1$, $s \le 4$	0/18
drop the $+j$ shift in m_j (Thm 14)	$r = 2$	0/8

Remark 18 (a half-dead control was the discovery instrument)**.** The first run of the sign-dropping control read 4/12 matching, not 0/12. Diagnosis (rather than hand-waving) showed all four "matches" were *degenerate samples*: the sign-flipped term was $\Psi^{\mathrm{C}}_{(1,1),(0,-1)}$, which is identically zero. That accident is what exposed Theorem 16, and $(0,-1)$ is the $r=2$ instance of $\theta_j-\theta_k=k-j$. Re-run on samples chosen off the vanishing wall, the control reads 0/20. My first guess at the locus — $\theta_1+\theta_2=-1$ — was *wrong*, and the datum that killed it was $\theta=(0,0,-2)$ at $r=3$, which vanishes with no pair difference equal to 1; the ρ-shift $m_j=\theta_j+j-1$ is what fits, 424/424.

7 T0: the Schur-Q kernel identification (retraction and result)

Retraction, 2026-10-23. The text of this section as first written graded a collaborator's note at commit `c53c581` as resting on an *unreadable source*, on the evidence that `git cat-file -t c53c581` returned `fatal: Not a valid object name` in all eight clones of his repository on this machine. **That grade was wrong and the hash is valid.** Measured on 2026-10-23 in both directions: `cat-file` returns `fatal` *before* `git fetch` and `commit` *after*; `gh api` resolves `c53c5813c1914ceca8d682b101d2e3e4f0b04777`, dated 2026-10-22T12:41:15Z. All eight clones shared a single stale fetch, so they were eight readings of one instrument and not eight instruments; counting clones is not replication, and the authority for a remote object is the remote. The collaborator's future-dated filename is his standing naming convention, not a tell, and that inference is withdrawn too.

The verdict was logged as **NO MATCH on unreadable source**, explicitly *not* on a measured mismatch. That is what makes this a retraction and not a withdrawal: no mathematics in this file depended on it.

And the identification holds. With the source now read, the collaborator's iterated residue formula couples its variables through $K(\tau,t)=(\tau-t)/(\tau t-1)$, and under his Cayley map $t=(1+z)/(1-z)$ this becomes the Schur-Q kernel $(z_r-z_s)/(z_r+z_s)$ (verified symbolically). Writing $\sigma_w(z)=\prod_{m\in S}(z-1/w_m)/(z-w_m)=P(z)/(e_S D(z))$ as in §3, the comparison is not an analogy but an identity:

$$\prod_{m\in S} K(z,w_m) \;=\; e_S\,\sigma_w(1/z) \;=\; \frac{1}{e_S\,\sigma_w(z)}.$$

Verified symbolically for $|S|\le 5$, with five controls all firing (dropping e_S, using e_S^2, the two wrong kernels $(a-b)/(ab+1)$ and $(a+b)/(ab-1)$, and omitting the reflection). So his coupling kernel, multiplied over the tail set, *is* the reflection operator of Theorem 10, and Theorem 10's functional equation $\sigma_w(1/z)=1/(e_S^2\sigma_w(z))$ is exactly the functional equation of his kernel product.

One correction in the other direction: the kernel is *not* antisymmetric under $t\mapsto 1/t$ in a single variable, as I had expected. It is reciprocal-valued there ($K\mapsto 1/K$) and invariant under the simultaneous inversion; its antisymmetry is under the *swap* $r\leftrightarrow s$, a different group element. Any argument that pairs the two involutions by their fixed points must use the simultaneous form.

8 Gaps and open points

1. **Converse of Theorem 16** — non-vanishing off the walls is `computed` (424/424), not proved; see Remark 17. A proof presumably needs a positivity or leading-term argument on $\sum_T B(T)s_\mu(T)$, and $B(T)$ is not of one sign.

2. **Generic clauses in Theorem 3** — the memberships of 0 for $t<s-1$ and of ∞ for $t\ge 1$ are coefficient non-vanishing statements, true off a proper subvariety; the set equality is verified

at 60 random rational points but is a generic, not universal, claim. The unconditional cases are $t = s - 1$, $t = 0$, and all w_m with $m \geq s$.

3. **Q553's counting question is answered in the negative but reopened in a better form.** There is no finite "number of admissible groupings"; the identity family is $\mathbb{Z}$-indexed with an explicit basis and an explicit closed form. Whether $\{I_a\}_{a \in \mathbb{Z}}$ is linearly independent *as functions of* w — which is the only sense in which a count is meaningful — is not settled here.

4. **T0** — a quoted peer artifact must be stamped with a commit hash that has been resolved by `git cat-file` *in the same action as the quote.* Here the hash was present and false. A locator written inside my own brief is not a locator: the quoting artifact and the quoted artifact were produced by the same slot minutes apart, so the quote's only corroboration was the brief that needed it.